Participant Profile Form

MSCA Staff Exchanges 2027 (MSCA-SE 2027) — consortium-building process

Unidad de Excelencia IOBA | University of Valladolid

Purpose:
This form is intended to collect structured information from interested participants for the preparation of an MSCA Staff Exchanges consortium. It is designed to support both (i) later completion of participant-related information requested in the EU Funding & Tenders Portal, and (ii) the creation of an internal profile database to help identify complementarities and facilitate contacts across the emerging consortium.

Please complete one form per:

- non-academic organisation (SMEs, NGOs , patient organisations)
- participating academic team (one form per PI, research-group, department).

This prospective **MSCA Staff Exchanges 2027 consortium** aims to build an international, interdisciplinary and inter-sectoral network in eye and vision research, centred on translational collaboration and staff mobility. Starting from Unidad de Excelencia IOBA's strengths in advanced therapies, biomarkers, inflammation, neuroprotection and health technology assessment, the network remains open to additional interdisciplinary collaborations, including areas such as ocular surface, retina, inherited and age-related eye disease, neuro-vision, optometry, drug delivery, AI-enabled diagnosis and patient-oriented validation, while remaining fully open to any other complementary expertise that could enrich the consortium and strengthen its patient-focused impact.

Important:
- please use this form to capture your general interests, expertise, and rough capacity to send and/or host staff.
- If you are not sure how to answer any field, you may leave it blank and highlight it.

1. Organisation and administrative profile

Organisation legal name	Linköpings Universitet
Short name / acronym used in projects	LIU
Participant Identification Code (PIC), if available	999852236
Website(s)	www.liu.se
Country	Sweden
Organisation type	☒ Academic ☐ Non-academic ☐ Hospital / clinic ☐ SME ☐ Large company ☐ Research infrastructure ☐ NGO / foundation ☐ Public body ☐ Other
Main address	Campus Valla, 581 83 Linköping, Sweden
Department(s), institute(s), research group(s) or unit(s) that would be involved	Department of Health, Medical and Caring Science, Linköping University, 581 83 Linköping, Sweden
Additional institutional resources relevant to collaboration	Pre-clinical: Zebrafish facilities and lab. Facilities for research using mouse, rat, rabbit and pig models. Facilities for molecular biology including qPCR, DNA/RNA sequencing, array analysis, proteomics and genome editing. Clinical: Advanced biobanks and clinical research units that support sample handling, recruitment of additional centers to multi-center studies, and oversight of clinical studies in all aspects from statistics, monitoration, legal and ethical aspects, and data management etc.

Main contact person for this profile	Lasse Jensen, Senior Associate Professor, PI, department of health, medical and caring sciences, division of diagnostics and specialist medicine, unit of cardiovascular medicine
Main contact email	Lasse.jensen@liu.se
Other contact person(s), if relevant	Lena Tasse, EU-project management support, lena.tasse@liu.se Rami Alzukimy, Economist, rami.alzukimy@liu.se

2. Scientific / innovation profile and relevance for an IOBA-led consortium

Please describe your profile at the level of the participating team / unit most relevant to the prospective consortium.

Indicative thematic relevance (tick all that apply)	
☐ Ocular surface	☒ Retina
☐ Inherited retinal dystrophies (IRDs)	☒ Age-related ocular disease
☐ Neuro-vision / integrated vision / neuro-ophthalmology	☐ Optometry / visual behaviour / rehabilitation
☐ Advanced therapies / regenerative medicine	☐ Gene therapy / cell therapy / tissue engineering
☒ Inflammation / biomarkers / neuroprotection	☐ Drug delivery / formulations / nanomedicine
☒ Drug discovery / screening / repurposing	☒ Ocular vascular disease
☐ Infection / trauma-related ocular disease	☐ Ocular pain / quality of life / patient-reported outcomes
☐ Health technology assessment	☐ AI-assisted diagnosis / software / digital tools
☐ Clinical trials / translational validation	☐ Other (please specify):

Free Keywords for internal searchable database (please provide up to 10)	Zebrafish; vasculature; blood-retinal barrier; choriocapillaris; edema; hypoxia
Geographies of interest for exchanges (optional)	

Overview of your research / innovation focus
We exploit embryonic, larval and adult zebrafish models of retinal or choroidal vascular dysregulation (induced mainly via exposure to hypoxia) to understand molecular and cellular mechanisms involved in vascular leakage, growth (both sprouting and intussusception), regression, effects on the surrounding tissue, and efficacy of newly identified targets or drug candidates. In addition to hypoxia, we have zebrafish models of proliferative diabetic retinopathy, inflammation- or neurodegeneration-induced vascular disruption, and vascular disruption associated to congenital eye diseases (e.g. FEVR, coloboma). *We are proficient in genetic engineering of zebrafish to create knockouts or knock-ins using Cas9:CRISPR. We are also routinely using regular or time-lapse confocal microscopy in longitudinal studies of vascular development or disruption under controlled conditions of retinal/sub-retinal disease progression. We have extensive experience in probing VEGF-pathways including an understanding of the importance of different VEGF-family members and receptors in different eye compartments.*

Why your team could be relevant to a translational eye-health and vision consortium
We can offer expertise and resources mainly in two areas of research: 1) Novel, innovative disease models and 2) Knowledge of and experience in vascular biology. *Zebrafish are becoming increasingly popular and relevant for eye-health and vision research due to their human-like, diurnal physiology (having both rods and cones, similar eye anatomy and molecular regulatory pathways), while also having increased regenerative capacity and pathophysiological relevance such as hypoxia- and diabetes-induced retinal angiogenesis and vascular*

disruption phenotypes associated with certain disease-related loss-of-function mutations. Furthermore, zebrafish embryos and larvae develop rapidly and are transparent which allows longitudinal visualization of any aspect of eye development, and the role of genes or drug in that process. Our experience allow both the development of new models to exploit the benefits of zebrafish in ophthalmology research and the use of our existing models for collaborative projects.

Vascular biology is studied also outside of the eye of the zebrafish (e.g. in cancer and in regulation of the blood-brain-barrier), as well as in other model organisms (the lab has extensive experience in mouse models). Blood vessels are key for many aspects of eye disorders, and the broad and deep knowledge in this area can benefit our collaborators to better study and understand the type of vascular disruption and its pathogenic roles.

3. Expertise, assets and collaboration interests

Core expertise, methods and platforms you could contribute

- *Zebrafish disease models*
- *Tools for and experience in studies of vascular biology*
- *Longitudinal imaging (e.g. time-lapse confocal microscopy)*
- *Microdissection (of zebrafish tissues including eyes)*
- *Knowledge of the innovation system including spin-off company creation and development*
- *Knowledge of and experience from translational studies incorporating patient material and information*

Infrastructure, equipment, facilities or datasets relevant to consortium work

- *Zebrafish facility and lab (including microinjection and microdissection systems)*
- *Cell culture labs*
- *Labs for molecular biology (qPCR, western blotting, sequencing, etc)*
- *Microscope facilities including various epi-fluorescence, confocal, light sheet, spinning disk, and two-photon microscopes*

Training capacity and knowledge-transfer potential

- *Experienced in training secondees on both short and long secondments*
- *Secondees will gain knowledge in zebrafish models, cell-based (in vitro) models, or associated techniques (microdissection, microinjection, cell culture, visualization)*
- *Secondees will gain knowledge in vascular biology both broadly (vessel development, structure, function, etc) and specifically how its coupled to the collaborative project*
- *Secondees will be in a translational environment highly used to closely collaboration with both clinicians and pharma/biotech companies*
- *Secondees will gain access to our tools and infrastructure, which can be exploited for adding data to ongoing projects*

What your team would hope to gain from participating

- *Access to complementary technologies such as cell- and/or mouse models or knowledge of how to analyze other aspects of zebrafish eye function)*
- *Complementary knowledge e.g. on how vascular disruption is affecting other parts of the eye and how vascular disruption can be avoided using innovative treatments*
- *Experience from other research environments which broaden the view on research for our early stage researchers*
- *Knowledge of and opportunities for alternative career-development paths for our early stage researchers e.g. at other European universities or in a non-academic sector.*

4. Key people likely to be involved

Please list the key people who may be relevant to the consortium at this stage. *This is not a final, binding or secondment list.*

Name	Position / role	Department / unit	Staff category	Career stage	Email / identifier (optional)
Jingdan Shen	Ph.D student	HMV/DISP	Researcher	D	Jingdan.shen@liu.se
Ashitha Vivekanandan	Ph.D student	HMV/DISP	Researcher	D	Ashitha.vadassery.vivekanandan@liu.se

Zhu Pengcheng	Ph.D student	HMV/DISP	Researcher	D	Zhu.pengcheng@liu.se

Main disciplines represented in your team	Vascular biology/pathology, zebrafish models, disease model engineering, fluorescence visualization, hypoxia studies
Experience in EU / international / multi-partner collaborative projects	Participation in three prior EU-staff-exchange programs, an EIC-Pathfinder consortium, two Eurostars consortia and as an early-stage researchers in ERC Advanced grant project and an FP-7 project. In addition, I have participated in six other, international multi-partner projects, of which two as the coordinator.
Experience with clinical, translational or patient-facing research (if relevant)	☒ Yes ☐ No ☐ Not applicable | Short note: Mainly within cancer

5. Evidence of capacity and track record

Up to 5 relevant publications / datasets / software / products / services / other achievements

*Lee C, Chen R, Sun G, Liu X, Lin X, He C, Xing L, Liu L, **Jensen L**, Kumar A, Langer H, Ren X, Zhang J, Huang L, Yin X, Kim J, Zhu J, Huang G, Li J, Lu W, Chen W, Liu J, Hu J, Sun O, Lu W, Fang L, Wang S, Kuang H, Zhang Y, Tian G, Mi J, Kang BA, Narazaki M, Prodeus A, Schoonjans L, Ornitz D, Gariepy J, Eelen G, Dewerchin M, Yang Y, Ou JS, Mora A, Yao J, Zhao C, Liu Y, Carmeliet P, Cao Y, and Li X, "VEGF-B prevents excessive angiogenesis by inhibiting FGF2/FGFR1 pathway", Signal Transduction and Targeted Therapy, 8(1):305*

*Karjosukarso DW, Ali Z, Peters TA, Zhang JQC, Hoogendoorn ADM, Garanto A, van Wijk E, **Jensen LD*** and Collin RWJ*, "Modeling ZNF408-associated FEVR in zebrafish results in abnormal retinal vasculature", Invest. Ophthalmol. Vis. Sci. 61(2):39*

*Ali Z, Zhang JJ, Lagali N, Schmitner N, Salvenmoser W, Mukwaya A, Neuhauss SCF, **Jensen LD*** and Kimmel RA*, "Photoreceptor degeneration accompanies vascular changes in a zebrafish model of diabetic retinopathy", Invest. Ophthalmol. Vis. Sci. 61(2):43*

*Ali Z, Mukwaya A, Biesemeier A, Ntzouni M, Ramsköld D, Giatrellis S, Mammadzada P, Cao R, Lennikov A, Marass M, Gerri C, Hildesjö C, Taylor M, Deng Q, Peebo B, del Peso L, Kvanta A, Sandberg R, Schraermeyer U, Andre H, Steffensen JF, Lagali N, Cao Y, Kele J and **Jensen LD** (2019), "Intussusceptive vascular remodeling precedes pathological neovascularization", Arterioscler. Throm. Vasc. Biol., 39(7), pp. 1402-18.*

*Ali Z, Yang Y, Tracey-White D, Vazquez-Rodriguez G, Moosajee M, Ju R, Li X, Cao Y and **Jensen LD** (2019), "Synchronized tissue-scale vasculogenesis and ubiquitous lateral sprouting underlie the unique architecture of the choriocapillaris", Dev. Biol. 457(2), pp. 206-14.*

Up to 5 relevant previous projects, grants, networks or collaborations

ALADDIN: EIC-PATHFINDER-OPEN project aiming to develop a new humanized zebrafish tumor xenograft platform for accelerated development of nanobody therapeutics in cancer (international)

EVEREST: MSCA-SE project wherein LIU is collaborating with 20+ EU partners to advance research in EV biology including applications within cancer research and novel therapies (international)

ROBO-FISH: A Eurostars project in which Dr. Jensen worked with LSM to develop new injection and visualization robots for creation and analysis of larval zebrafish tumor xenograft models. (international)

MECHANO-CHIP: A Eurostars project in which Dr. Jensen was collaborating with Dutch partners to develop microfluidic systems for cancer research (international)

Swedish Research Council: A highly competitive Sweden-China collaboration grant to in collaboration with Prof. Xuri Li at Sun Yat-Sen Ophthalmic Center in Guangzhou, China, establish shared capabilities for research on the role of PDGF-C and VEGF-B in ophthalmology

Patents, protected know-how, clinical protocols, validated tools or other translational assets (if any)

Please indicate only what can be shared at this stage.

6. Staff-exchange capacity (only rough estimation at this stage)

Please provide only a rough, non-binding estimate of your organisation's/team's capacity to send and/or host staff **over the 48-month duration of the project.**

Could your organisation/team second staff?	☒ Yes ☐ Maybe ☐ No capacity to send staff on secondment
Could your organisation second staff categories beyond researchers (where relevant to R&I activities)?	☐ Only Researchers ☒ Technical Staff/Innovation ☒ Managerial / project support ☐ Admin Staff ☐ Other profiles
Could your organisation host incoming staff from academic and/or non-academic sectors?	☐ Academic only ☐ Non-academic only ☒ Both ☐ No capacity to host
Administrative / HR / legal considerations that may affect mobility	Periodically difficult to get independent access to facilities (due to delays in administration), Proficiency in English language is required.

7. Matchmaking information: Summary for the internal participant database

Capabilities you would most like other interested participants to notice
Experts in eye vascular biology *Zebrafish models for eye diseases (focus on the vasculature)* *Translational research profile and experience working with pre-clinical methodologies in clinical studies/contexts*

Capabilities or expertise you would especially like to access through the consortium
OCT-A capabilities and knowledge *Clinical samples and/or multi-OMICs data from patients with eye diseases driven by vascular disruption (e.g. PDR, AMD, FEVR, etc)* *Mouse models of PDR and/or AMD to validate findings from zebrafish models*

Please Confirm:

Open to being contacted by other interested participants through the coordination team's mediation?	☒ Yes ☐ No
☒ We understand that the information contained in this form will be searchable and accessible to other interested participants within the pre-consortium process.	

8. Final remarks and query items

Please note any uncertainties, prerequisites, internal approvals needed, or items that should be considered further before the consortium is consolidated.
Examples: PIC pending approval, institutional approval, legal status, hosting rules, IP constraints, data access, ethics, clinical permissions, staffing continuity, resource availability.

Please provide the following details (as many entries as needed).

Internal use and confidentiality statement

The information provided in this form will be used only for internal consortium-building purposes in connection with the preparation of a prospective MSCA Staff Exchanges proposal (2027 call) coordinated from Unidad de Excelencia IOBA / University of Valladolid. It may be used by the coordination team to: (i) assess strategic fit and administrative readiness, (ii) prepare participant-related information required at proposal stage, and (iii) build an internal profile database to facilitate contacts among interested participants. The information will not be shared outside this pre-consortium context without prior permission from the contributing organisation / contact person, except where later required for the formal preparation of a proposal with the explicit involvement of that organisation. Submission of this form confirms that the contributor agrees to this internal use. Organisations may request correction, update or withdrawal of their information at any time by contacting the coordination team.

☐ **Understood**

Name:

Date completed:

(Coordination team: ivan.fernandez.bueno@uva.es; yolanda.alvarez.teruel@uva.es)